Engineering Portfolio

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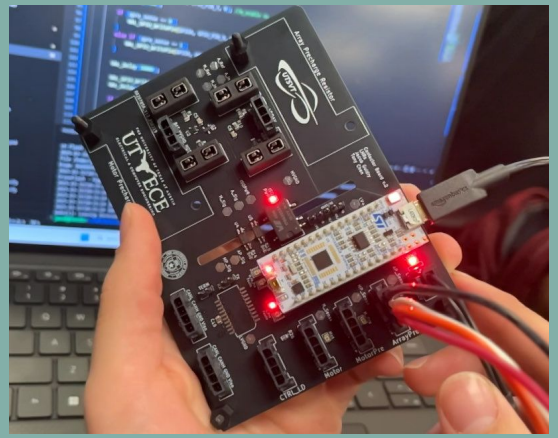
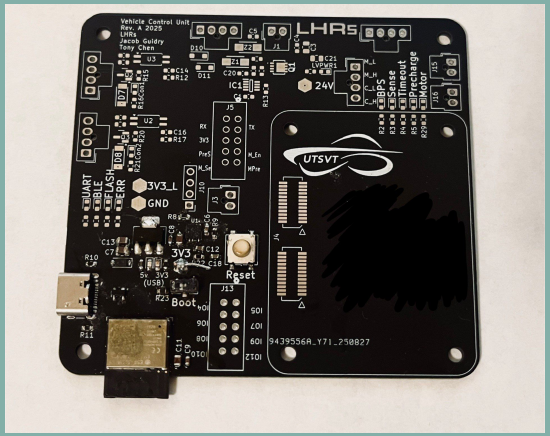
Longhorn Racing Solar

As the High Voltage Lead for UT's Longhorn Racing Solar Team, I have developed multiple systems and led team projects involving PCB design, embedded software, system validation and debugging, and much more.

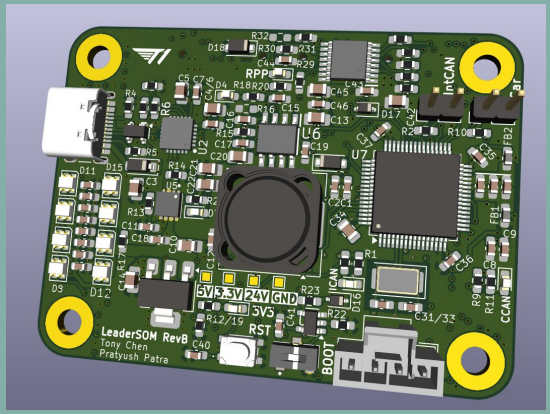
Active-Precharge Board



Vehicle Control Unit



Leader System-on-Module



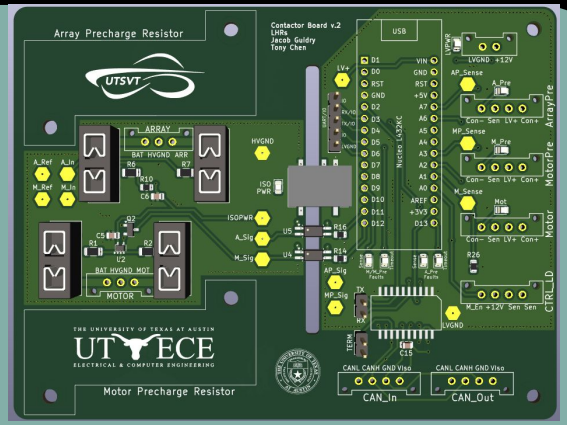
Active-Precharge Board

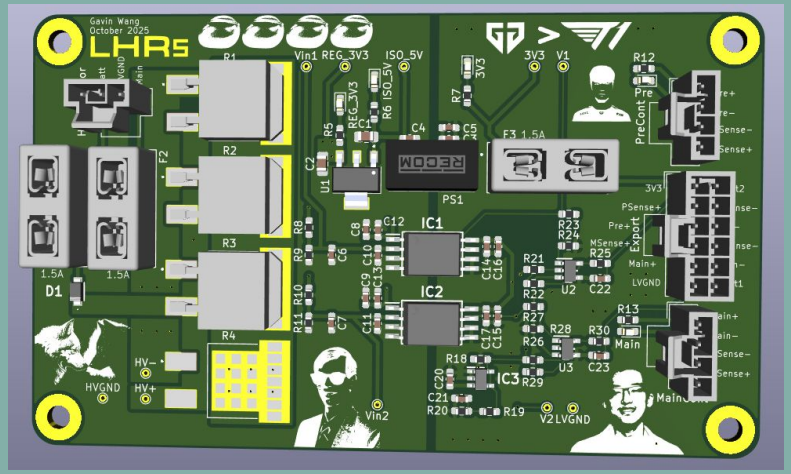
Problem: Precharge process to energize the motor controller and solar array was slow and unsafe. “Dumb” precharge simply waited a set period of time for capacitors to charge through a large resistor.

Problem Solution: Intelligent “Active” precharge directly measures the high voltage lines to determine when the motor controller or solar array have been energized to the same voltage as the battery. Additionally, rapid prototyping and redesign was necessitated due to frequent scope changes.

Right: Version I developed and used at summer 2025 competition

Bottom: Latest revision designed by a new member I trained

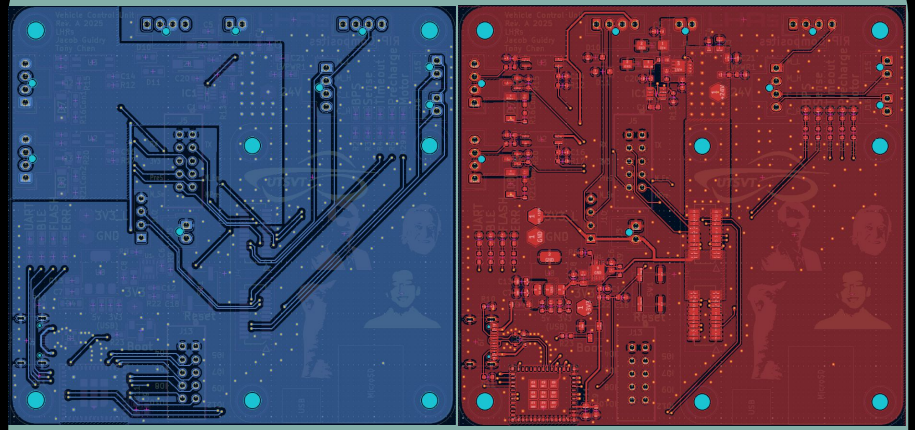




Vehicle Control Unit (VCU)

Central vehicle motor controller communication hub and high voltage safety management board.

- Split communication across multiple different CANs. CarCAN for central controls and MotorCAN for motor controller messages
- Designed smart circuitry to control motor power relays and precharge state.
- Wireless flashing using ESP32 microcontroller
- Power protection circuitry for reverse polarity, over current, and over voltage detection.



Top: Top and Bottom Layers of PCB design

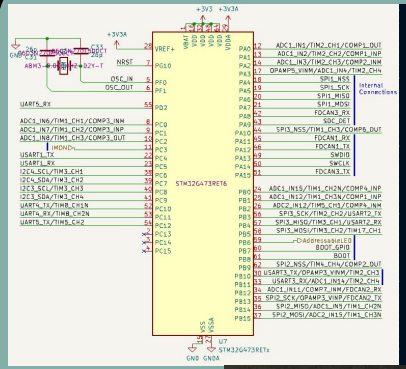
Right: Partially assembled version being tested

Leader System on Module (LSOM)

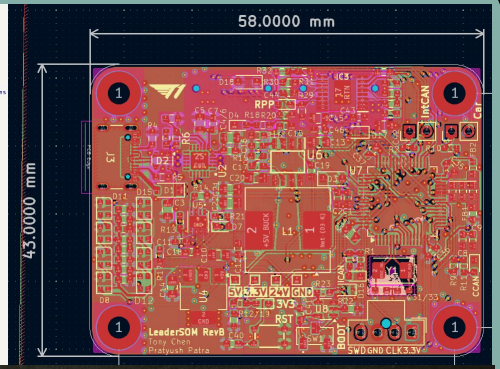
The **Leader System-On-Module (LSOM)** is an embedded hardware platform I designed as a control and communication module for multiple subsystems on the Longhorn Racing Solar vehicle. The board integrates an STM32G473 to support CAN, USB, UART, SPI, and I²C communication, and provides regulated power distribution. Designed as a 4-layer PCB, the LSOM emphasizes signal integrity, power efficiency, and robustness in high-noise automotive environments.

I led the design from schematic capture through PCB layout, firmware, and validation, delivering a reusable module that was completed in a rapid development cycle and deployed across several critical vehicle systems.

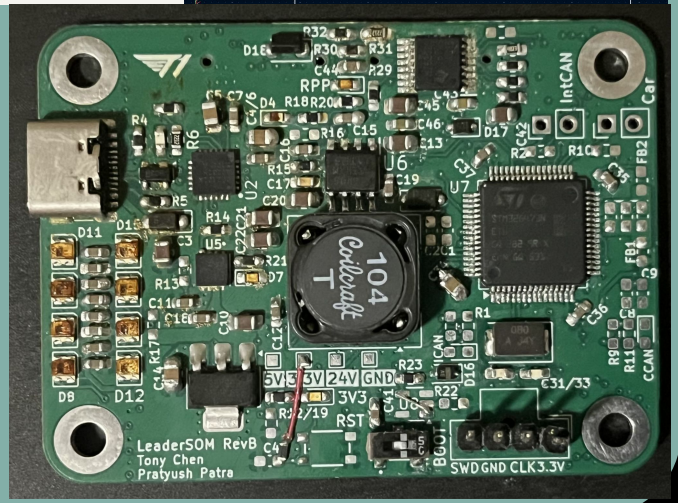
Microcontroller Pinout



Circuit Board Layout

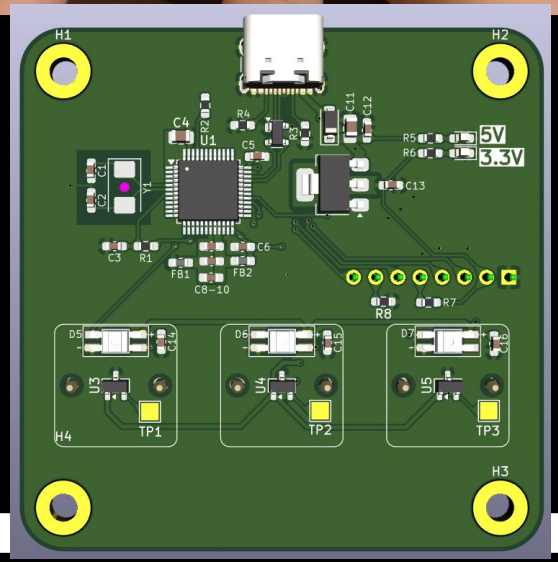


Debugging and testing prototype



ECE 319K Microcontroller Project (Tetris)

Custom Analog Keyboard



Personal/Academic Projects

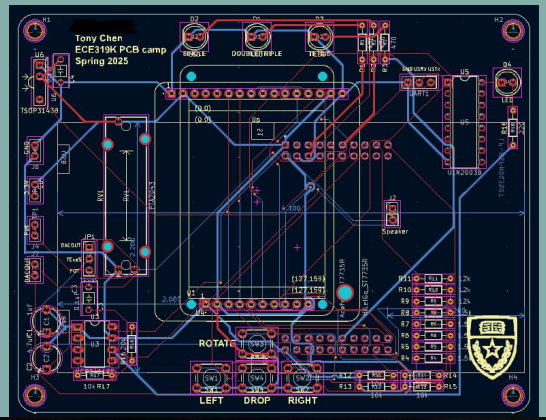
ECE 319K Microcontroller Project (Tetris)

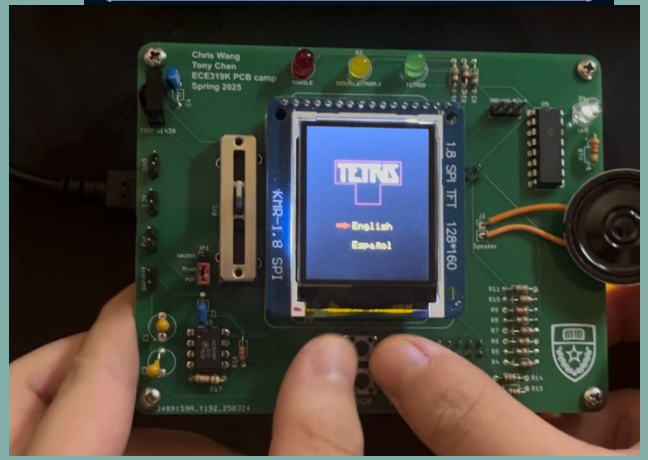
For my ECE 319K Embedded Systems final lab project, I designed and built a handheld gaming device that played Tetris in 3 weeks. It used the TI MSPM0 microcontroller to integrate an LCD display, push-button inputs, and a modern guideline tetris game engine implemented in C and C++. I optimized both hardware interfacing and software timing to deliver responsive visuals and efficient gameplay.

My project placed in the top 5 of the class competition, demonstrating strong performance, creativity, and attention to embedded design principles.

Right: Circuit Layout designed in 1 week with classmate partner

Bottom: [Game running](#) on MSPM0 Microcontroller



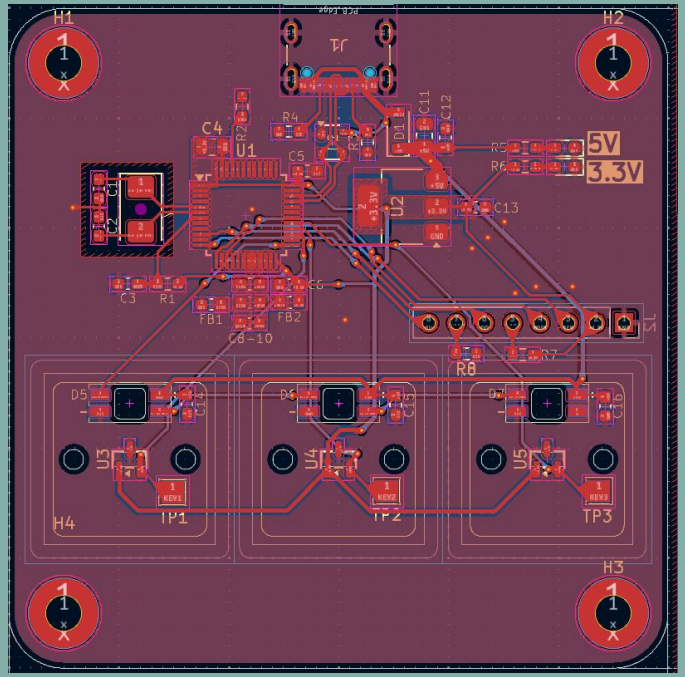


Custom Analog Keyboard

I am developing a custom analog keypad that uses linear Hall Effect/TMR magnetic sensors to detect key position. Instead of mechanical switches, each key's motion is tracked by monitoring the magnetic field from a magnet, enabling smooth analog input. I designed the complete system, including the sensor interface, analog signal conditioning, and microcontroller firmware. Future plans are to build a full sized keyboard based on the same principles.

This project highlights my experience with custom analog PCB design, embedded firmware development, and prototyping novel user-interface hardware

Underside of PCB featuring STM32G491 microcontroller, SK6812MINI-E LEDs, and DRV5055A Hall Effect Sensors



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THANK YOU!